

Absolutely. Below is a **complete project-planning document** that consolidates the bookstore, semantic recommendation system, AI chatbot, roles, stakeholders, features, architecture, database, APIs, security, and 7-day implementation plan.

AI-Powered MERN Book Store — Complete Project Plan

1. Project Overview

Project Name

AI-Powered Book Store Management and Recommendation System

Technology

Frontend

- React.js
- React Router
- Axios
- Tailwind CSS
- React Hook Form

Backend

- Node.js
- Express.js
- JWT
- bcrypt
- REST APIs

Database

- MongoDB
- Mongoose

AI

- Existing Semantic Book Recommendation System
 - Embedding-based semantic search
 - LLM-powered chatbot
 - RAG
 - Tool/function calling
-

2. Project Vision

The system is an online bookstore that combines traditional e-commerce functionality with AI.

A customer can:

Discover Books



Search Books



Get Semantic Recommendations



Chat with AI Book Assistant



Add Books to Cart



Checkout



Place Order



Track Order



Review Books

The key differentiator is that the system doesn't simply display books. It **understands what the customer is looking for** and provides personalized assistance.

3. Major System Modules

AI BOOKSTORE



— Authentication & Authorization

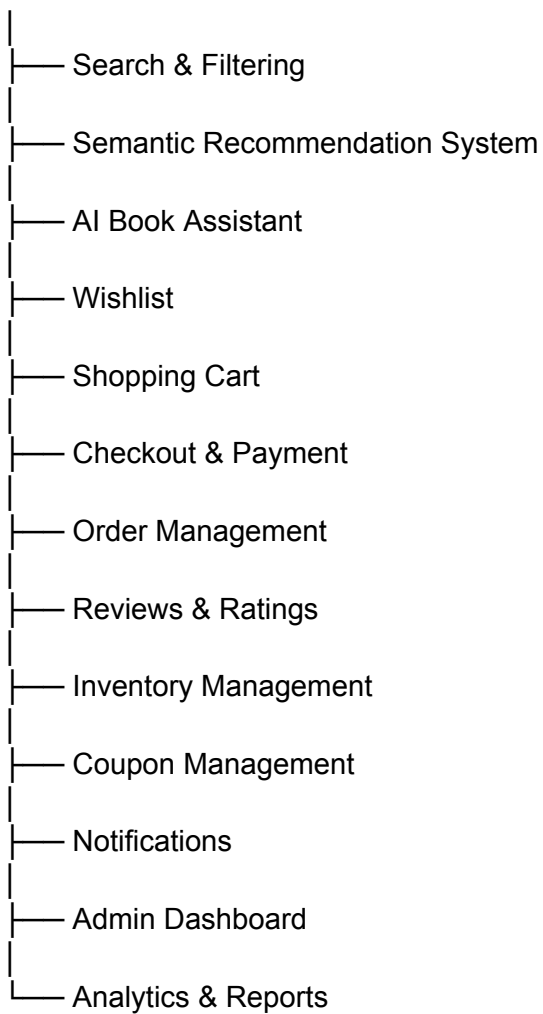
— User Management

— Book Management

— Category Management

— Author Management

— Publisher Management



4. System Actors / Roles

Keep the actual application RBAC simple:

1. Guest

Unauthenticated visitor.

2. Customer

Registered user who purchases and interacts with the system.

3. Admin

Full system-management user.

AI Recommendation Engine

System component responsible for semantic recommendations.

AI Book Assistant

System component that interacts with users and invokes authorized backend tools.

5. Stakeholders

Stakeholder	Main Interest
Customer	Find and purchase books
Guest	Discover books
Admin	Manage bookstore
Store Owner	Sales and business analytics
Book Manager	Book/catalog management
Order Manager	Order fulfillment
Author	Book visibility
Publisher	Book distribution
Delivery Partner	Delivery information
Payment Provider	Payment processing
Developer	Development and maintenance
AI Recommendation Engine	Personalized discovery
AI Chatbot	Customer assistance

6. Guest Features

Guests can:

- View homepage
- Browse books
- View book details
- Search books

- Semantic search
- Filter books
- Sort books
- Browse categories
- View authors
- View publishers
- View ratings
- Read reviews
- View similar books
- View trending books
- Use general AI chatbot
- Register
- Login

They cannot:

- Purchase
 - Checkout
 - Maintain a cart
 - Use personalized recommendations
 - Write reviews
 - Manage wishlist
 - View orders
-

7. Customer Features

Account

- Register
- Login
- Logout
- Email verification
- Forgot password
- Reset password
- Change password
- Edit profile
- Profile picture
- Manage addresses

Book Discovery

- Browse books
- Search

- Semantic search
- Filter
- Sort
- View book details
- View authors
- View publishers
- View categories
- View reviews

Shopping

- Add to cart
- Remove from cart
- Update quantity
- Checkout
- Apply coupons
- Place order
- View order history
- Track orders
- Cancel eligible orders
- View invoice

Engagement

- Wishlist
- Ratings
- Reviews
- Recently viewed

AI

- Personalized recommendations
- Similar books
- Conversational recommendations
- Book comparison
- Natural-language book search
- AI chatbot
- Order assistance
- Cart assistance
- Wishlist assistance

8. Admin Features

Dashboard

- Total users
- Total books
- Total orders
- Revenue
- Pending orders
- Completed orders
- Low-stock books
- Out-of-stock books
- Best-selling books
- Most-viewed books
- Popular categories

User Management

- View users
- Search users
- Edit users
- Activate/deactivate users
- Delete users
- Manage roles

Book Management

- Add
- Edit
- Delete
- View
- Search
- Upload cover
- Update price
- Update stock
- Manage descriptions
- ISBN
- Author
- Category
- Publisher
- Language
- Edition
- Pages

Catalog Management

- Categories

- Authors
- Publishers

Order Management

- View all orders
- Search/filter orders
- Update status
- Process orders
- Ship orders
- Mark delivered
- Cancel orders

Inventory

- Stock management
- Restocking
- Low-stock alerts
- Out-of-stock monitoring

Reviews

- View reviews
- Delete inappropriate reviews
- Monitor ratings

Coupons

- Create
 - Edit
 - Delete
 - Activate/deactivate
 - Expiration
 - Usage limits
-

9. Semantic Recommendation System

Your existing recommendation system becomes an integrated AI service.

Main functions

Similar Book Recommendation

Book



Embedding



Similarity Search



Top-K Books

Displayed as:

You May Also Like

Personalized Recommendation

Uses:

- Viewed books
- Purchased books
- Wishlist
- Search history
- Ratings
- Favorite genres

Output:

Recommended For You

Semantic Search

Example:

"I want an easy book for learning Python."

The system understands the **meaning** of the query rather than requiring exact keywords.

10. AI Chatbot — Book Assistant

The chatbot should be an actual **AI agent for the bookstore**, not merely an FAQ bot.

General Questions

- Store information
- Book information
- Author information
- Publisher information
- Categories
- Shipping
- Payment
- Returns
- Refunds
- Policies

Book Search

User:

"Find me beginner programming books under ₺800."

The chatbot translates this into structured search criteria and calls your backend.

11. Chatbot Recommendation Features

User:

"I liked Clean Code. What should I read next?"

Chatbot:

User preference



Semantic recommendation engine



Relevant books



LLM



Natural-language response

It can combine:

- Semantic similarity
- Rating

- Price
 - Availability
 - User history
-

12. Chatbot Book Comparison

User:

"Compare these two books."

The chatbot can compare:

- Topic
 - Author
 - Price
 - Rating
 - Pages
 - Level
 - Availability
 - Description
-

13. Chatbot Shopping Assistant

The chatbot can:

- View cart
 - Add book to cart
 - Remove book
 - Change quantity
 - Calculate cart total
 - View wishlist
 - Add to wishlist
 - Remove from wishlist
-

14. Chatbot Order Assistant

Customer can ask:

"Where is my latest order?"

"Show my orders."

"What did I order last month?"

"Can I cancel my order?"

The chatbot calls authenticated backend APIs.

Important:

LLM must never directly modify MongoDB.

Use:

LLM

↓

Tool Calling

↓

Backend Authorization

↓

Business Logic

↓

MongoDB

15. Chatbot Tools

Read tools

searchBooks()

getBookDetails()

semanticSearch()

getSimilarBooks()

getRecommendations()

getOrderStatus()

getOrderHistory()

getCart()

getWishlist()

searchFAQ()

Write tools

addToCart()

removeFromCart()

```
updateCart()  
addToWishlist()  
removeFromWishlist()  
cancelOrder()
```

Write operations require backend authorization and, where appropriate, user confirmation.

16. RAG System

For store policies and FAQ:

```
graph TD  
    A[User Question] --> B[Embedding]  
    B --> C[Vector Search]  
    C --> D[Relevant Documents]  
    D --> E[LLM]  
    E --> F[Answer]
```

Knowledge base:

```
graph TD  
    A[FAQ] --- B[Shipping]  
    A --- C[Payment]  
    A --- D[Refund]  
    A --- E[Return]  
    A --- F[Ordering]  
    A --- G[Account]  
    A --- H[Store Policies]
```

This is preferable to simply putting all policies into the chatbot's prompt.

17. Database Design

User

```
{  
  name,  
  email,  
  passwordHash,  
  phone,  
  role,  
  profileImage,  
  
  wishlist: [],  
  viewHistory: [],  
  searchHistory: [],  
  favoriteGenres: [],  
  
  addresses: [],  
  
  createdAt,  
  updatedAt  
}
```

Book

```
{  
  title,  
  author,  
  description,  
  category,  
  publisher,  
  
  isbn,  
  language,  
  edition,  
  pages,  
  
  price,  
  stock,  
  
  coverImage,  
  
  embedding: [],  
  
  averageRating,  
  reviewCount,  
  
  viewCount,
```

```
purchaseCount,  
  
tags: [],  
  
createdAt,  
updatedAt  
}
```

Order

```
{  
  user,  
  items: [  
    {  
      book,  
      quantity,  
      price  
    }  
  ],  
  
  shippingAddress,  
  
  paymentMethod,  
  paymentStatus,  
  
  subtotal,  
  discount,  
  total,  
  
  status,  
  
  createdAt,  
  updatedAt  
}
```

Review

```
{  
  user,  
  book,  
  rating,  
  comment,  
  createdAt,
```

```
    updatedAt
  }
```

Category

```
{
  name,
  description,
  createdAt
}
```

Author

```
{
  name,
  biography,
  image,
  createdAt
}
```

Publisher

```
{
  name,
  description,
  website,
  createdAt
}
```

Coupon

```
{
  code,
  type,
  value,
  minimumPurchase,
  expiryDate,
  usageLimit,
  active
}
```



```
}
```

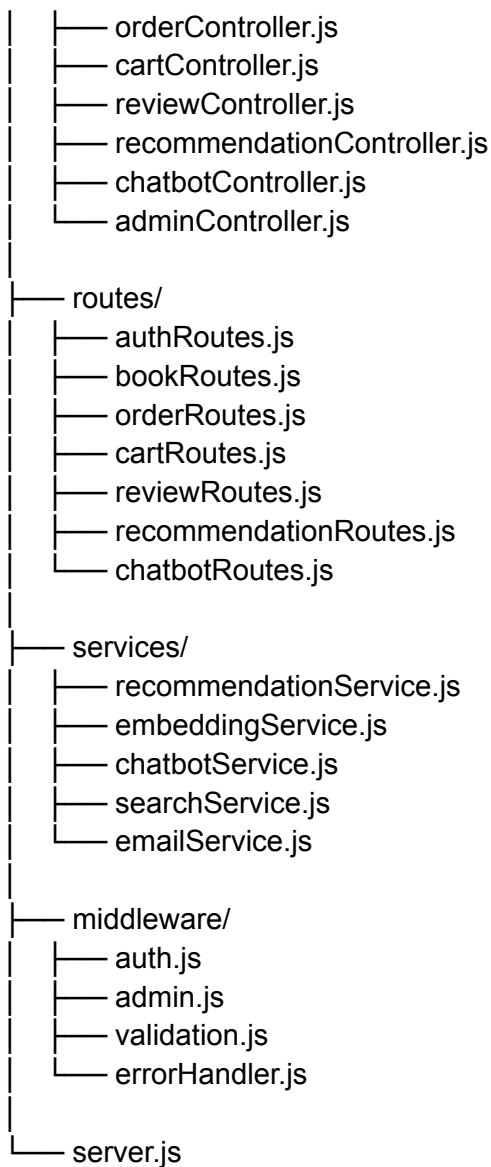
Chat Conversation

```
{  
  user,  
  messages: [  
    {  
      role,  
      content,  
      timestamp  
    }  
  ],  
  
  createdAt,  
  updatedAt  
}
```

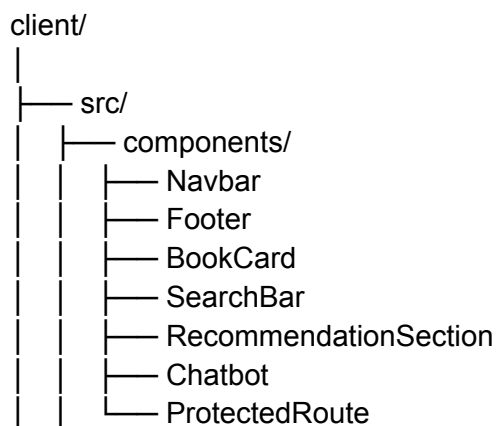
Don't store sensitive information unnecessarily.

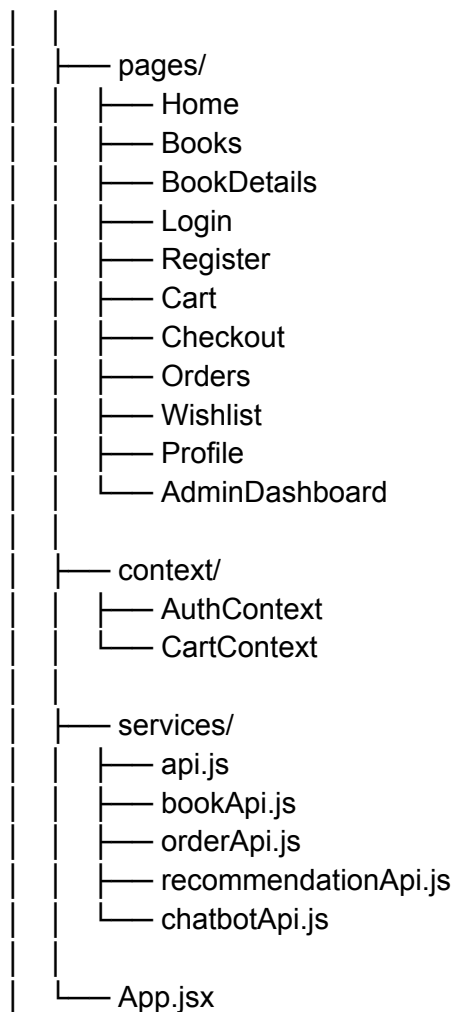
18. Backend Architecture

```
server/  
├── config/  
│   ├── db.js  
│   └── ai.js  
├── models/  
│   ├── User.js  
│   ├── Book.js  
│   ├── Order.js  
│   ├── Review.js  
│   ├── Category.js  
│   ├── Author.js  
│   ├── Publisher.js  
│   ├── Coupon.js  
│   └── Conversation.js  
├── controllers/  
│   ├── authController.js  
│   └── bookController.js
```



19. Frontend Architecture





20. API Structure

Authentication

POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
GET /api/auth/profile
PUT /api/auth/profile

Books

GET /api/books
GET /api/books/:id
POST /api/books
PUT /api/books/:id

DELETE /api/books/:id

Search

GET /api/search

GET /api/search/semantic

Recommendations

GET /api/recommendations/similar/:bookId

GET /api/recommendations/personalized

GET /api/recommendations/trending

Cart

GET /api/cart

POST /api/cart

PUT /api/cart/:bookId

DELETE /api/cart/:bookId

Orders

POST /api/orders

GET /api/orders

GET /api/orders/:id

PUT /api/orders/:id/cancel

Reviews

GET /api/books/:id/reviews

POST /api/books/:id/reviews

PUT /api/reviews/:id

DELETE /api/reviews/:id

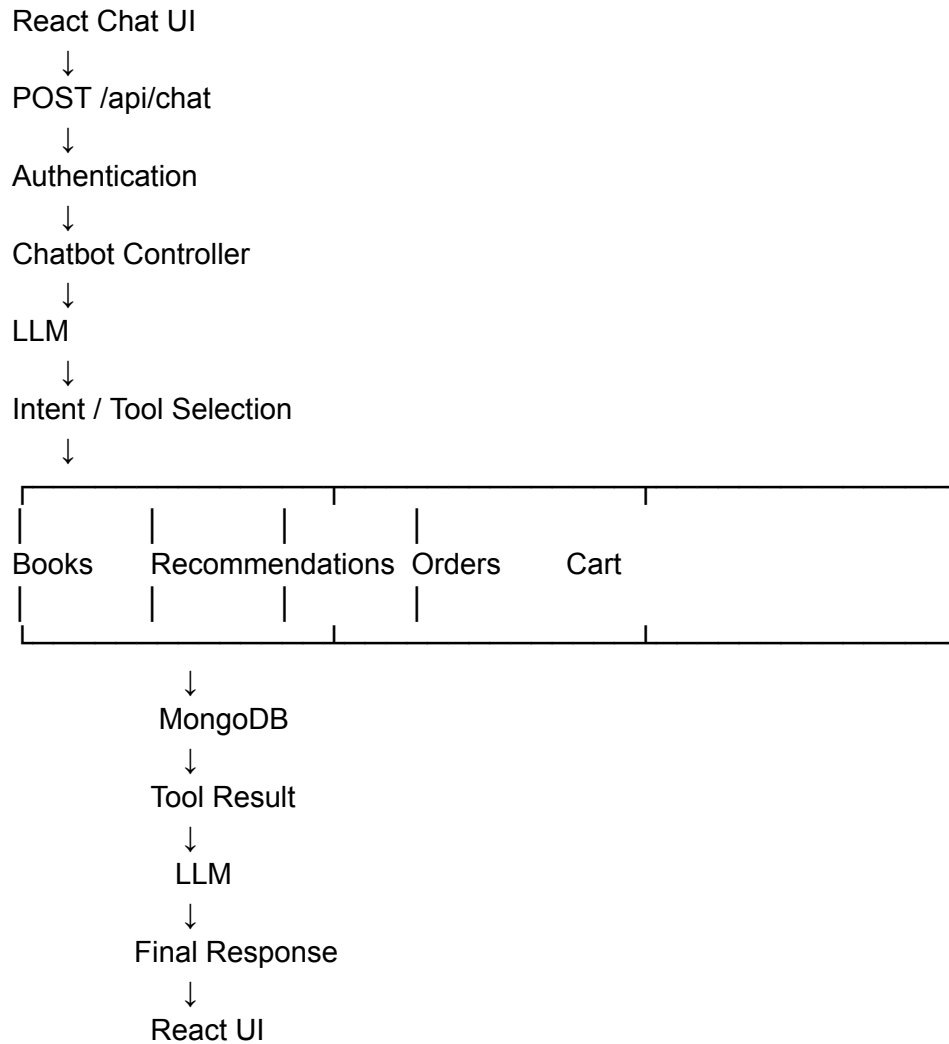
Chatbot

POST /api/chat

GET /api/chat/history

DELETE /api/chat/history

21. Chatbot API Flow



22. Security Architecture

Implement:

- JWT authentication
- bcrypt password hashing
- Role-based authorization
- Protected API routes
- Input validation
- Rate limiting
- CORS
- Secure HTTP headers
- Environment variables
- File-upload validation

- MongoDB authorization
- API error handling

AI-specific security

The chatbot must:

- Never receive raw database credentials
- Never directly execute arbitrary database queries
- Never bypass authorization
- Never expose another user's data
- Validate all tool arguments
- Require confirmation for sensitive actions
- Restrict tools based on user role

For example:

Customer
↓
"Show admin sales data"
↓
Authorization check
↓
DENIED

23. Use Cases

Guest

Register
Login
Browse Books
Search Books
Semantic Search
View Book
View Categories
View Author
View Reviews
View Similar Books
Use AI Assistant

Customer

Manage Profile
Manage Wishlist

Manage Cart
Search Books
Semantic Search
Get Recommendations
Chat with AI
Compare Books
Place Order
Checkout
Make Payment
View Orders
Track Order
Cancel Order
Review Book
Rate Book
Manage Addresses
Apply Coupon

Admin

Manage Users
Manage Books
Manage Categories
Manage Authors
Manage Publishers
Manage Inventory
Manage Orders
Manage Reviews
Manage Coupons
View Analytics
Manage Recommendations
View AI Analytics

24. Complete Access Matrix

Feature	Guest	Customer	Admin
Browse	✓	✓	✓
Search	✓	✓	✓
Semantic Search	✓	✓	✓
Similar Books	✓	✓	✓

Personalized Recommendations	✗	✓	—
AI Chatbot	Basic	Full	Admin Assistant
Wishlist	✗	✓	✗
Cart	✗	✓	✗
Checkout	✗	✓	✗
Orders	✗	Own	All
Reviews	Read	Own	All
Books	Read	Read	CRUD
Categories	Read	Read	CRUD
Authors	Read	Read	CRUD
Inventory	✗	✗	✓
Users	✗	Own	CRUD
Coupons	Apply	Apply	CRUD
Sales Analytics	✗	✗	✓
Recommendation Analytics	✗	✗	✓
System Settings	✗	✗	✓

25. Seven-Day Development Plan

Day 1 — Foundation

Backend

- Create Express project
- MongoDB connection
- Mongoose setup
- User model
- JWT
- bcrypt
- Authentication

- Role middleware

Frontend

- React/Vite
- Tailwind
- Router
- Axios
- Authentication context
- Navbar
- Login/Register

Goal

Authentication + project foundation complete.

Day 2 — Books & Catalog

Backend

- Book model
- Category
- Author
- Publisher
- CRUD APIs
- Search
- Filtering
- Sorting
- Pagination

Frontend

- Homepage
- Book catalog
- Book card
- Book details
- Search
- Filters
- Categories

Goal

Complete bookstore catalog.

Day 3 — E-Commerce

Backend

- Cart
- Orders
- Checkout
- Inventory
- Coupon
- Order status

Frontend

- Cart
- Checkout
- Order confirmation
- Orders page
- Order details

Goal

Customer can purchase books.

Day 4 — Recommendation System

Integrate your existing semantic recommendation system.

Implement

- Embedding generation
- Embedding storage
- Semantic search
- Similar books
- Personalized recommendations
- Trending books
- Recently viewed

Frontend

- Recommended section
- Similar books

- Recommended for You
- Semantic search UI

Goal

AI recommendation system integrated.

Day 5 — Reviews + Wishlist + Admin

Customer

- Wishlist
- Reviews
- Ratings
- Profile
- Addresses

Admin

- Dashboard
- Book CRUD
- Category CRUD
- Author CRUD
- Publisher CRUD
- User management
- Inventory
- Orders

Goal

Complete core bookstore functionality.

Day 6 — AI Chatbot

Implement:

Chat UI

- Floating chatbot
- Message history
- Loading indicator

- Error handling
- Book cards inside responses

AI

- LLM integration
- Conversation context
- RAG
- Book search
- Semantic search
- Recommendation tools
- FAQ tools

Tools

searchBooks()
getBookDetails()
getRecommendations()
getSimilarBooks()
getOrderStatus()
getCart()
getWishlist()
addToCart()
addToWishlist()

Goal

AI Book Assistant fully functional.

Day 7 — Polish, Testing & Deployment

Testing

- Authentication
- Authorization
- Book CRUD
- Cart
- Orders
- Reviews
- Recommendations
- Chatbot
- Tool permissions

UI

- Responsive design
- Loading states
- Error states
- Empty states
- Toast notifications

Deployment

React



Vercel

Node/Express



Render / Railway

MongoDB



MongoDB Atlas

Documentation

- README
 - Architecture
 - ER/database diagram
 - Use-case diagram
 - API documentation
 - Screenshots
 - Deployment instructions
-

26. Priority Levels

Because you only have **7 days**, don't try to implement everything simultaneously.

Must Have

Authentication

Book CRUD

Search

Categories

Cart

Orders

Reviews

Wishlist

Admin
Semantic Recommendations
AI Chatbot

● Should Have

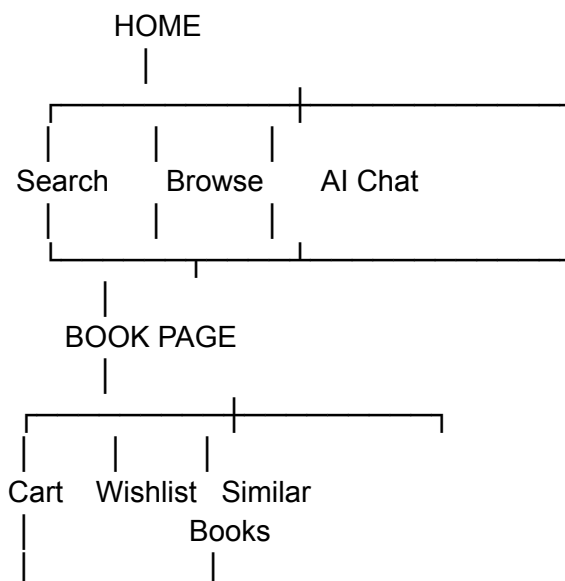
Coupons
Analytics
Recently Viewed
Order Tracking
FAQ/RAG
Book Comparison
Recommendation Analytics

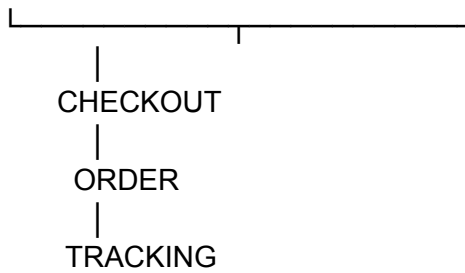
● Nice to Have

Voice Chat
Real-time notifications
Advanced payment integration
AI recommendation explanations
Conversational checkout
Sales forecasting
Advanced AI analytics

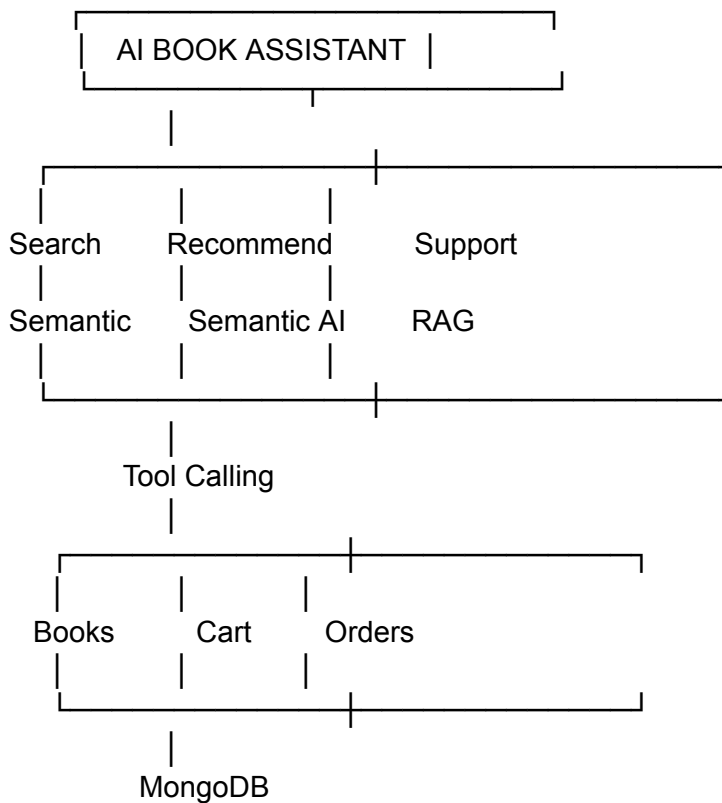
27. Final Product

At the end of the project, the user experience should look like:





And alongside the normal bookstore:



Final Project Scope

Your final system can be presented as:

A full-stack AI-powered e-commerce bookstore built with the MERN stack, featuring semantic book search, personalized book recommendations, intelligent conversational assistance, secure role-based access control, shopping cart and order management, reviews, wishlist, inventory management, and administrative analytics.

That gives you a strong **software engineering + MERN + AI** project rather than simply a CRUD bookstore.